

A Hierarchical Bayesian Framework for Self-Consistent Sea Level Rise Forecasting

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Abstract

We present a hierarchical Bayesian framework for probabilistic sea level rise projections that integrates observational constraints, process-model structural information, and deep-uncertainty treatment into a single self-consistent system. The framework operates across four levels: (0) global temperature forcing, (1) total GMSL constrained by the observational rate–temperature relationship, (2) component-level decomposition into seven physically distinct contributions (thermal expansion, glaciers, Greenland, West Antarctic, East Antarctic, Antarctic Peninsula, terrestrial water storage) with component-specific forcing variables and a rate-level budget constraint, and (3) deep-uncertainty modules for marine ice-sheet instability. Three features distinguish the framework from existing approaches. First, a rate-level soft budget constraint enforces the sea level budget at every calibration time step, creating cross-component information flow via a collider mechanism that transfers information from well-observed components to poorly observed ones. Second, the directed acyclic graph (DAG) specification supports causal intervention analysis via do-calculus, enabling counterfactual projections under glacier stabilization or climate engineering scenarios. Third, learned Kennedy & O’Hagan model-inadequacy terms at each component level prevent the overconfidence that arises from treating parametric models as truth. The framework is designed for incremental implementation: it is self-consistent at every stage of completeness, with unresolved components absorbed into a budget-constrained residual. We demonstrate the framework at Stage 1 (thermостeric decomposition) and Stage 2 (glaciers and Greenland), producing projections that are observationally calibrated, budget-consistent, and have explicit model-inadequacy accounting.

1 Introduction

2 Framework Architecture

2.1 Hierarchical structure

2.2 Directed acyclic graph

2.3 Rate-level budget constraint

2.4 Model inadequacy

2.5 Staged implementation

3 Demonstration: Stages 1-2

- 3.1 Stage 1: Thermosteric decomposition
- 3.2 Stage 2: Glaciers and Greenland
- 3.3 Projection comparison
- 3.4 Model inadequacy diagnosis
- 4 Causal Intervention Analysis
- 5 Variance Decomposition
- 6 Relationship to Existing Work
- 7 Discussion
- 8 Conclusions